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GATE ECE · ELECTROMAGNETICS · CHAPTER-WISE PYQS

Maxwell Equations

GATE ECE Electromagnetics · Previous Year Questions with Answer Key

48	32	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	25 questions
Marks 2	23 questions
Answer key	at the end

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Marks 1 25 questions

Q1. GATE ECE 2026

Which option(s) represents/ represent the dielectric loss tangent of a substrate?

(A) Ratio of the real to imaginary parts of the total displacement current

(B) $\frac{(\omega\epsilon'' + \sigma)}{(\omega\epsilon')}$

(C) Ratio of the electric susceptibility to permittivity

(D) Ratio of the polarization vector $\vec{P}$ to the displacement vector $\vec{D}$

Open solution

Q2. GATE ECE 2024

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x and y axes, respectively and let A be a positive constant. Which one of the following statements is true for the vector fields $\vec{F}_1 = A(\hat{i}y + \hat{j}x)$ and $\vec{F}_2 = A(\hat{i}y - \hat{j}x)$?

(A) Both $\vec{F}_1$ and $\vec{F}_2$ are electrostatic fields.

(B) Only $\vec{F}_1$ is an electrostatic field.

(C) Only $\vec{F}_2$ is an electrostatic field.

(D) Neither $\vec{F}_1$ nor $\vec{F}_2$ is an electrostatic field.

Open solution

Q3. GATE ECE 2022

In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is 2sin(t + π/2). The displacement current (in Amperes) through the capacitor is _____.

(A) 2sin(t) (B) 2sin(t + π) (C) 2sin(t + π/2) (D) 0

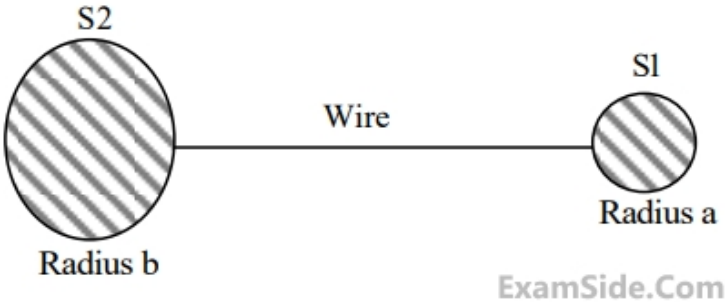
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Q4. GATE ECE 2017 Set 2

Two conducting spheres S1 and S2 of radii a and b (b>a) respectively, are placed far apart and connected by a long, thin conducting wire, as shown in the figure.

Open solution



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For some charge placed on this structure, the potential and surface electric field on S1 are V_a and E_a , and that on S2 are V_b and E_b, respectively, which of the following is CORRECT?

(A) $V_a = V_b$ and $E_a < E_b$

(B) $V_a > V_b$ and $E_a > E_b$

(C) $V_a = V_b$ and $E_a > E_b$

(D) $V_a > V_b$ and $E_a = E_b$

Q5. GATE ECE 2016 Set 2

A uniform and constant magnetic field $B = \hat{z}B$ exists in the $\hat{z}$ direction in vacuum. A particle of mass m with a small charge q is introduced into this region with an initial velocity $v = \hat{x}v_x + \hat{z}v_z$. Given that B, m, q, v_x and v_z are all non-zero, which one of the following describes the eventual trajectory of the particle?

Open solution

(A) Helical motion in the $\hat{z}$ direction.

(B) Circular motion in the xy plane.

(C) Linear motion in the $\hat{z}$ direction.

(D) Linear motion in the $\hat{x}$ direction.

Q6. GATE ECE 2016 Set 2

Consider the time-varying vector $I = \hat{x}\ 15\ \cos(\omega t) + \hat{y}5\ sin(\omega t)$ in Cartesian coordinates, where ω > 0 is a constant. When the vector magnitude $|I|$ is at its minimum value, the angle θ that I makes with the x axis (in degree, such that $0 \leq 0 \leq 180$) is _____

Open solution

Q7. GATE ECE 2016 Set 1

Concentric spherical shells of radii 2 m, 4 m, and 8 m carry uniform surface charge densities of 20 nC/m^2 , −4 nC/m^2 and ρ_s ,respectively. The value of ρ_s (nC/m^2) required to ensure that the electric flux density $\overrightarrow{D} = \overrightarrow{0}$ at radius 10 m is _____.

Open solution

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Q8. GATE ECE 2016 Set 3

Faraday's law of electromagnetic induction is mathematically described by which one of the following equations?

(A) $\nabla \bullet \vec{B} = 0$

(B) $\nabla \bullet \vec{D} = \rho_v$

(C) $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

(D) $\nabla \times \vec{H} = \sigma\vec{E} + \frac{\partial \vec{D}}{\partial t}$

Open solution

Q9. GATE ECE 2015 Set 2

In a source free region in vacuum, if the electrostatic potential $\varphi = 2x^2 + y^2 + cz^2$, the value of constant c must be _____.

Open solution

Q10. GATE ECE 2013

The divergence of the vector field $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is

(A) 0 (B) 1/3 (C) 1 (D) 3

Open solution

Q11. GATE ECE 2011

Consider a closed surface S surrounding volume V. If $\vec{r}$ is the position vector of a point inside S, with $\hat{n}$ the unit normal on S, the value of the integral is

$$\oint_S 5\vec{r} \cdot \hat{n}\, dS$$

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(A) 3 V (B) 5 V (C) 10 V (D) 15 V

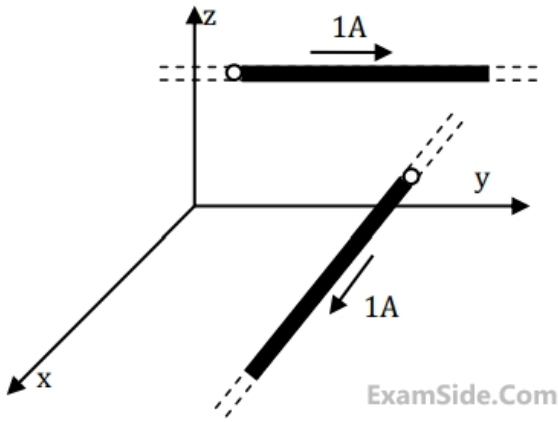
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Q12. GATE ECE 2009

Two infinitely long wires carrying current are as shown in the figure below. One wire is in the y-z plane and parallel to the y-axis. The other wire is in the x-y plane and parallel to the x-axis. Which components of the resulting magnetic field are non-zero at the origin?

Open solution



(A) x, y, z components

(B) x, y components

(C) y, z components

(D) x, z components

Q13. GATE ECE 2007

If C is a closed curve enclosing a surface S, then the magnetic field intensity $\vec{H}$, the current density $\vec{J}$ and the electric flux density $\vec{D}$ are related by

Open solution

(A) $\iint_S \vec{H}.\vec{ds} = \oint_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right).\vec{dl}$

(B) $$\int_C \vec{H}.\vec{dl} = \iint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right).\vec{ds}$$

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(C) $$\iint_S \vec{H}.\vec{ds} = \int_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right).\vec{dl}$$

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(D) $\oint_C \vec{H}.\vec{dl} = \iint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right).\vec{ds}$

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Q14. GATE ECE 2006

$\int \int (\nabla \times \mathrm{P}) \cdot \mathrm{ds}$, where is a vector, is equal to

(A) $\mathrm{P} \times \nabla \times \mathrm{P} - \nabla^2 \mathrm{P}$

(B) $\nabla^2 \mathrm{P} + \nabla (\nabla \cdot \mathrm{P})$

(C) $\nabla^2 \mathrm{P} + \nabla \times \mathrm{P}$

(D) $\nabla (\nabla \cdot \mathrm{P}) - \nabla^2 \mathrm{P}$

Open solution

Q15. GATE ECE 2006

$\nabla \times \nabla \times P$, where P is a vector, is equal to

(A) $P \times \nabla \times P - \nabla^2 P$

(B) $\nabla^2 P + \nabla (\nabla . P)$

(C) $\nabla^2 P + \nabla \times P$

(D) $\nabla (\nabla . P) - \nabla^2 P$

Open solution

Q16. GATE ECE 2003

The unit of $\nabla \times \mathrm{H}$ is

(A) Ampere

(B) Ampere/meter

(C) Ampere/meter^2

(D) Ampere - meter

Open solution

Q17. GATE ECE 1999

An electric field on a plane is described by its potential V = $20\left(r^{-1} + r^{-2}\right)$ where r is the distance from the source. The field is due to

(A) a monopole

(B) a dipole

(C) both a monopole and a dipole

(D) a quadrupole

Open solution

Q18. GATE ECE 1999

Identify which one of the following will **NOT** satisfy the wave equation.

(A) $50e^{j(\omega t - 3z)}$

(B) $\sin[\omega(10z + 5t)]$

(C) $\cos\left(y^2 + 5t\right)$

(D) $\sin(x).\cos(t)$

Open solution

Q19. GATE ECE 1998

The Maxwell's equation $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ is based on

(A) Ampere's law

(B) Gauss' law

(C) Faraday's law

(D) Coulomb's law

Open solution

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Q20. GATE ECE 1998

A loop is rotating about the y-axis in a magnetic field

$$\overrightarrow{B} = B_0 \cos(\omega t + \phi)\overrightarrow{a_x}\ \mathrm{T}.$$

The voltage in the loop is

(A) zero

(B) due to rotation only

(C) due to transformer action only

(D) due to both rotation and transformer action

Open solution

Q21. GATE ECE 1998

The time averaged Poynting vector, in W/m^2, for a wave with $\vec{E} = 24e^{j(\omega t+\beta z)}\overrightarrow{a_y}$ V/m in free space is

(A) $-\frac{2.4}{\pi}\overrightarrow{a_z}$

(B) $\frac{2.4}{\pi}\overrightarrow{a_z}$

(C) $\frac{4.8}{\pi}\overrightarrow{a_z}$

(D) $-\frac{4.8}{\pi}\overrightarrow{a_z}$

Open solution

Q22. GATE ECE 1996

A metal sphere with 1 m radius and a surface charge density of 10 Coulombs/m^2 is enclosed in a cube of 10 m side. The total outward electric displacement normal to the surface of the cube is

(A) 40π Coulombs

(B) 10π Coulombs

(C) 5π Coulombs

(D) None of the above

Open solution

Q23. GATE ECE 1995

The electric field strength at a distance point, P, due to a point charge, +q, located at the origin, is 100 μV/m. If the point charge is now enclosed by a perfectly conducting metal sheet sphere whose center is at the origin, then the electric field strength at the point P outside the sphere becomes.

(A) Zero

(B) 100 μV/m

(C) -100 μV/m

(D) 50 μV/m

Open solution

Q24. GATE ECE 1995

In the infinite plane, y = 6 m, there exists a uniform surface charge density of (1/6000π) μC/m^2. The associated electric field strength is:

(A) $30\,\hat{i}$ V/m

(B) $3\,\hat{j}$ V/m

(C) $30\,\hat{k}$ V/m

(D) $60\,\hat{j}$ V/m

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Q25. GATE ECE 1994

$$\oint_C \vec{A}.d\vec{\ell} = \int_S _____ .d\,\vec{s}$$

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Open solution

Marks 2 23 questions

Q26. GATE ECE 2025

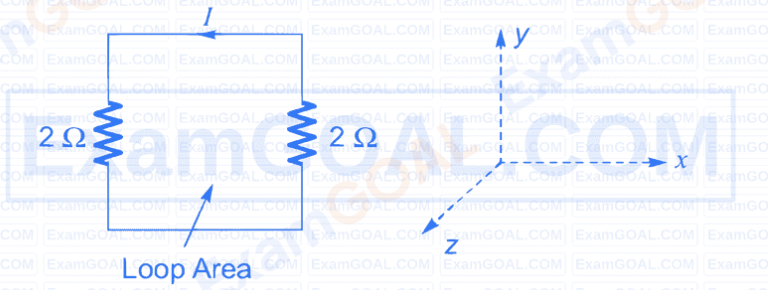
An electric field of $0.01\ \mathrm{V/m}$ is applied along the length of a copper wire of circular crosssection with diameter 1 mm . Copper has a conductivity of $5.8 \times 10^7\ \mathrm{S/m}$. The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____

(A) 0.46 (B) 1.82 (C) 0.58 (D) 1.12

Open solution

Q27. GATE ECE 2025

Two resistors are connected in a circuit loop of area $5\ \mathrm{m}^2$, as shown in the figure below. The circuit loop is placed on the $x-y$ plane. When a time-varying magnetic flux, with flux-density $B(t) = 0.5t$ (in Tesla), is applied along the positive z-axis, the magnitude of current I (in Amperes, rounded off to two decimal places) in the loop is_____ .



Open solution

Q28. GATE ECE 2022

In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by

$\vec{D}(x, y, z) = (x^3\vec{i} + y^3\vec{j} + xy^2\vec{k})$ C/m^2,

where, $\vec{i}, \vec{j}, \vec{k}$ are the unit vectors along x-axis, y-axis, and z-axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at ($\pm$ 0.5 m, $\pm$ 0.5 m, $\pm$ 0.5 m). The electric charge enclosed within R is _____ C (rounded off to two decimal places).

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Q29. GATE ECE 2017 Set 2

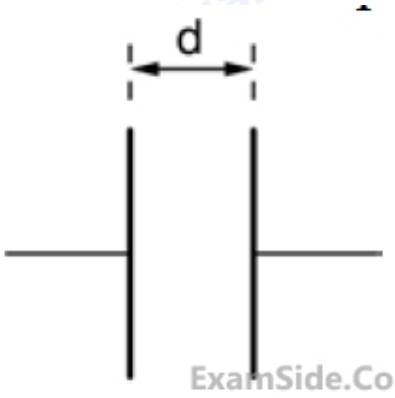
An electron (q_1) is moving in free space with velocity 10^5 m/s towards a stationary electron (q_2) far away. The closest distance that this moving electron gets to the stationary electron before the repulsive force diverts its path is _____ $\times 10^{-8}$m.

[Given, mass of electron m = 9.11×10^{-31} kg, charge of electron e = -1.6×10^{-19} C , and permittivity $\varepsilon_0 = (1/36\pi) \times 10^{-9}$ F/m].

Open solution

Q30. GATE ECE 2016 Set 2

The parallel-plate capacitor shown in the figure has movable plates. The capacitor is charged so that the energy stored in it is E when the plate separation is d. The capacitor is then isolated electrically and the plates are moved such that the plate separation becomes 2d.



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Open solution

At this new plate separation, what is the energy stored in the capacitor, neglecting fringing effects?

(A) 2E (B) $\sqrt{2}$E (C) E (D) E/2

Q31. GATE ECE 2015 Set 3

A coaxial capacitor of inner radius 1 mm and outer radius 5 mm has a capacitance per unit length of 172 pF/m. If the ratio of outer radius to inner is doubled, the capacitance per unit length (in pF/m) is _____.

Open solution

Q32. GATE ECE 2014 Set 4

If $\vec{E} = -(2y^3 - 3yz^2)\hat{x} - (6xy^2 - 3xz^2)\hat{y} + (6xyz)\hat{z}$
is the electric field in a source free region, a valid expression for the electrostatic potential is

(A) $xy^3 - yz^2$ (B) $2xy^3 - xyz^2$ (C) $y^3 + xyz^2$ (D) $2xy^3 - 3xyz^2$

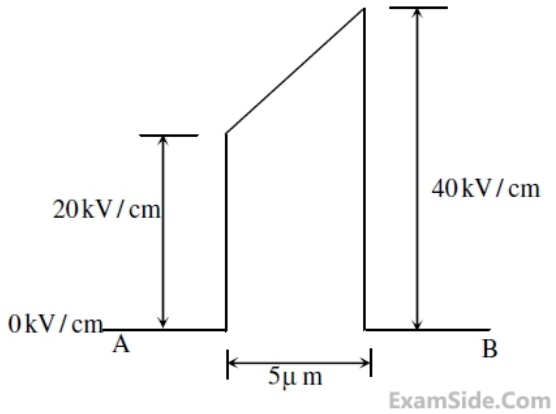
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Q33. GATE ECE 2014 Set 4

Open solution

The electric field (assumed to be one-dimensional) between two points A and B is shown. Let ψ_A and ψ_B be the electrostatic potentials at A and B, respectively. The value of $\psi_A - \psi_B$ in Volts is _____.



Q34. GATE ECE 2014 Set 3

Open solution

Given the vector

$$A = (\cos\ x)\,(\sin\ y)\,\hat{a}_x + (\sin\ x)\,(\cos\ y)\hat{a}_y$$

where $\hat{a}_x$, $\hat{a}_y$ denote unit vectors along x, y directions, respectively. The magnitude of curl of A is _____

Q35. GATE ECE 2012

Open solution

An infinitely long uniform solid wire of radius a carries a uniform dc current of density $\hat{j}$
The magnetic field at a distance r from the center of the wire is proportional to

(A) r for r < a and $1/r^2$ for r > a

(B) 0 for r < a and 1/r for r > a

(C) r for r < a and 1/r for r > a

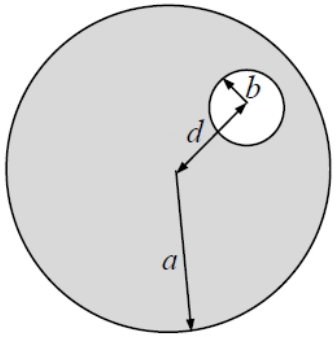
(D) 0 for r < a and $1/r^2$ for r > a

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Q36. GATE ECE 2012

Open solution

An infinitely long uniform solid wire of radius a carries a uniform dc current of density $\hat{j}$.
A hole of radius b (b < a) is now drilled along the length of the wire at a distance d from the center of the wire as shown below.



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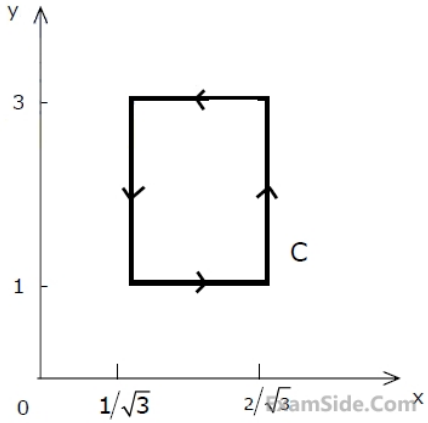
The magnetic field inside the hole is

(A) uniform and depends only on d
(B) uniform and depends only on b
(C) uniform and depends on both b and d
(D) non uniform

Q37. GATE ECE 2010

Open solution

If $\overrightarrow{A} = xy\,\hat{a}_x + x^2\,\hat{a}_y$ then $\oint \overrightarrow{A}.\overrightarrow{dl}$ over the path shown in the figure is



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(A) 0
(B) $\frac{2}{\sqrt{3}}$
(C) 1
(D) $2\sqrt{3}$

Q38. GATE ECE 2008

Open solution

The value of the integral of the function $g(x, y) = 4x^3 + 10y^4$ along the straight line segment from the point (0, 0) to the point (1, 2) in the x-y plane is

(A) 33
(B) 35
(C) 40
(D) 56

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Q39. GATE ECE 2003

If the electric field intensity is given by $\mathrm{E} = \left(\mathrm{x}\hat{\mathrm{u}}_{\mathrm{x}} + \mathrm{y}\hat{\mathrm{u}}_{\mathrm{y}} + \mathrm{z}\hat{\mathrm{u}}_{\mathrm{z}}\right)\ \mathrm{volt/m}$, the potential difference between X(2,0,0) and Y(1,2,3) is

(A) +1 volt (B) -1 volt (C) +5 volt (D) +6 volt

Open solution

Q40. GATE ECE 2001

A material has conductivity of 10^{-2} mho/m and a relative permittivity of 4. The frequency at which the conduction current in the medium is equal to the displacement current is

(A) 45 MHz (B) 90 MHz (C) 450 MHz (D) 900 MHz

Open solution

Q41. GATE ECE 1993

Given, $\overrightarrow{V} = x\cos^2 y\ \hat{\mathrm{i}} + \mathrm{x}^2\mathrm{e}^{\mathrm{z}}\ \hat{\mathrm{j}} + \mathrm{z}\sin^2\mathrm{y}\ \hat{\mathrm{k}}$ and S the surface of a unit cube with one corner at the origin and edges parallel to the coordinate axes, the value of the integral is

$$\iint_S \vec{V}.\hat{\mathrm{n}}\ dS$$

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Open solution

Q42. GATE ECE 1990

The incoming solar radiation at a place on the surface of the earth is 1.2 KW/m^2.The amplitude of the electric field corresponding to this incident power is nearly equal to

(A) 80 mV/m (B) 2.5 V/m (C) 30 V/m (D) 950 V/m

Open solution

Q43. GATE ECE 1990

Which of the following field equations indicate that the free magnetic charges do not exits?

(A) $\overrightarrow{H} = \frac{1}{\mu}\nabla \times A$

(B) $\overrightarrow{H} = \oint \frac{Id\ell \times R}{4\ \pi\ R^2}$

(C) $\nabla . \overrightarrow{H} = 0$

(D) $\nabla \times \overrightarrow{H} = J$

Open solution

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Q44. GATE ECE 1989

The electric field strength at a far-off point P due to a point charge + q, located at the origin O is 100 millivolts/metre. The point charge is now enclosed by a perfectly conducting hollow metal sphere with its centre at the origin O. The electric field strength at the point, P,

(A) remains unchanged in its magnitude and direction

(B) remains unchanged in its magnitude but reverse in direction

(C) would be that due to a dipole formed by the charge + q at O and - q induced

(D) would be zero

Open solution

Q45. GATE ECE 1988

Vector potential is a vector

(A) whose curl is equal to the magnetic flux density

(B) whose curl is equal to the electric field intensity

(C) whose divergence is equal to the electric potential

(D) which is equal to the vector product E x H

Open solution

Q46. GATE ECE 1988

On either side of a charge-free interface between two media,

(A) the normal components of the electric field are equal

(B) the tangential components of the electric field are equal

(C) the normal components of the electric flux density are equal

(D) the tangential components of the electric flux density are equal

Open solution

Q47. GATE ECE 1987

An electrostatic field is said to be conservative when

(A) The divergence of the field is equal to zero.

(B) The curl of the field is equal to zero.

(C) The curl of the field is equal to $-\frac{\partial B}{\partial t}$.

(D) The Laplacian of the field is equal to $\mu \varepsilon \frac{\partial^2 E}{\partial t^2}$.

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Q48. GATE ECE 1987

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Two long parallel wires in free space are separated by a distance R and carry currents of equal magnitude but opposite in direction. At any general point, the Z-component of

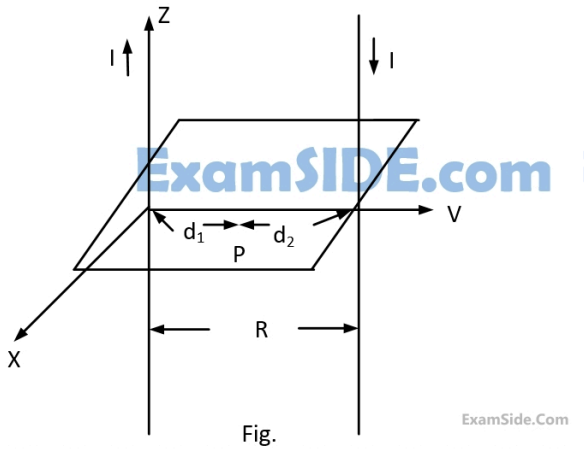


Fig.

(A) The magnetic vector potential is $\frac{\mu_o I}{4\pi} \ell n \left(\frac{{d_2}^2}{{d_1}^2}\right)$

(B) The magnetic induction is $\frac{\mu_o I}{2\pi} \left(\frac{d_2}{d_1}\right)$

(C) The magnetic induction is zero

(D) The magnetic vector potential is $\frac{\mu_o I}{2\pi} \left(\frac{{d_2}^2}{{d_1}^2}\right)$

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Answer Key

Marks 1

1. **A, B**	2. **B**	3. **C**	4. **C**	5. **A**
6. **90**	7. **-0.25**	8. **C**	9. **-3**	10. **D**
11. **D**	12. **D**	13. **D**	14. **A**	15. **D**
16. **C**	17. **C**	18. **C**	19. **A**	20. **D**
21. **A**	22. **A**	23. **A**	24. **B**	25. $\nabla \times \vec{A}$

Marks 2

26. **A**	27. **0.62 to 0.63**	28. **0.5**	29. **5.06**	30. **A**
31. **120.22**	32. **D**	33. **-15**	34. **0**	35. **C**
36. **A**	37. **C**	38. **A**	39. **C**	40. **A**
41. **(1/2)sin(2)**	42. **D**	43. **C**	44. **D**	45. **A**
46. **B, C**	47. **B**	48. **A**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
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